

TECHNICAL DEEP DIVE

How the system works

Architecture, data model, and the rules that decide what a visit costs — including the ones that were wrong first.

For whoever maintains this next

Shape of it

```
Browser (Next.js 14, App Router, React)
|
|  supabase-js over HTTPS
v
Supabase – Postgres + Auth + Storage
|
+-- Row Level Security: the real access boundary
```

There is no server of our own. The browser talks to Postgres directly through PostgREST, and **Row Level Security** decides what it may see. That is deliberate: a UI check can be bypassed with developer tools, a database policy cannot.

Area	Choice	Why
Framework	Next.js 14, App Router	Marketing pages and app in one deployment
Styling	Tailwind over CSS variables	Brand colour changes at runtime, not build time
Data	Supabase (Postgres)	Auth, storage and RLS without a backend to run
Payments	Square Point of Sale API	Readers cannot be driven remotely; the app hands off
Email	Resend, via a server route	Keeps the API key off the client

Route groups

The public site and the application share one deployment but not one identity. `app/(site)/` is the marketing website; everything else is the product. The split is listed once, in `lib/routes.ts`, because several behaviours depend on it — dark mode is staff-only, and the runtime page title must not overwrite the SEO titles the marketing pages set.

Data model

Table	Holds	Note
<code>dogs</code>	Every dog, with the full enrollment questionnaire	Keyed to a household by phone
<code>owners</code>	Contact, emergency and vet details	One per phone number
<code>signins</code>	One row per drop-off and per pick-up	Not one row per visit — see below
<code>boardings</code>	Reservations	One dog each; multi-dog requests fan out

<code>packages</code> / <code>package_uses</code>	Blocks bought, and each day spent	The ledger is what makes deduction auditable
<code>walk_logs</code> / <code>meal_logs</code>	Care during a stay	Walk rows now name the dog directly
<code>enrollments</code> / <code>boarding_requests</code>	Submitted forms awaiting review	The raw form is kept as JSONB
<code>payments</code>	Money received	Per household, not per dog
<code>settings</code>	One row. The whole configuration	What makes it not one-business-only

Why sign-ins are two rows

A visit is a drop-off row and, later, a pick-up row. A single row with two timestamps would have to be updated on the way out, and a boarding stay's pick-up can be days after its drop-off. Two rows also survive the awkward cases: a dog collected by someone else, a correction, an evening drop-off spanning midnight.

The pairing happens in `buildOpenVisits()` : last drop-off after the last pick-up means the dog is here. Every screen asks that one function, so the board, the kiosk and the profile cannot disagree about who is on site.

Pricing

Every rate is read live from `settings` through getters, so pricing functions stay pure and synchronous while remaining editable at runtime.

What a package covers

Situation	Result
Daycare over 4 hours, block available	Spends a day automatically; base rate free
Daycare under 4 hours, nobody chose a block	Nothing spent; billed as a half day
Daycare under 4 hours, staff chose a block	Spends a day; billed at zero, no half-day fee
Boarding	Never touches a package

The difference between the middle two is whether staff made a choice, which the code now carries explicitly as `pkgExplicit` . Inferring it from "a package exists" was wrong: it silently spent a whole day on a two-hour visit.

Walk packages are daycare-only

Boarding walks bill per walk on the reservation. Letting a stay also draw from a walk block produced a genuine double-spend — the stay consumed its block *and* the pick-up consumed one more.

Spending a day exactly once

Staff can assign a package from the list before the dog leaves, which writes a ledger row immediately so the count on screen is true. Sign-out would then deduct a second day. `adoptExistingUse()` looks for an unattached use of the same *kind* from today and links it to this visit instead of consuming another. Matching on kind rather than on the exact block matters: if staff earmarked block A and the default resolves to block B, the visit still owes one day, and staff's choice is the one that should stand.

Time

Postgres `current_date` is UTC. The business is in Pacific time. Seeded data landed a day in the future until every date was computed in local time. Anywhere a day is derived from a timestamp, it goes through `dateKey()` — never `created_at.slice(0, 10)`, which is the UTC day and files an evening drop-off under tomorrow.

Theming

Staff pick one brand colour. The eleven-step ramp is derived by mixing toward white and black, and the text colour on top is chosen by WCAG contrast — so a pale brand colour gets dark text without anyone deciding. Everything paints through CSS custom properties, which is why the colour can change at runtime.

Signed in as **frontdesk**

Sign out
Kiosk

Dashboard
In House
Calendar
Packages
Requests
5
Day report
Settings



Buki
Alvarez
(415) 555-0132
[Parent/Guardian's profile](#)
[Change photo](#)

No record
\$250.00 owed
9 of 10 walks left
7 of 10 days left
In since 7:42 AM

HANDLING NOTES

- Allergies: Chicken

▼ BASIC INFO
Hide

Dog name
Owner last name
Usual drop-off / pick-up

Buki
Alvarez
Ana Alvarez

Breed
Sex
Spayed / neutered

Border Collie
Female
Spayed

Date of birth - 5 yr 3 mo
Weight (lb)
Colour

2021-04-12
38
Black & white

Flea program
Where they came from
Vet (if different from the owner's)

Not answered
Breeder, shelter, rescue...
Leave blank to use the household v

Also authorized to pick up

Anyone else allowed to collect this dog

Phone number is the owner's — change it on the [owner profile](#).

▶ HEALTH & GROOMING
No health problems · Allergies: Chicken
Show

▶ HISTORY
Nothing recorded
Show

▶ BEHAVIOUR & PLAY
Friendly, Energetic · big dogs: does well · Daycare often
Show

▼ VACCINES
Hide

VACCINE	DATE GIVEN	EXPIRES	STATUS
Rabies	2025-03-04	2028-03-04	Up to date

Default.

Signed in as **frontdesk** Sign out ← Kiosk

[Dashboard](#)
[In House](#)
[Calendar](#)
[Packages](#)
[Requests](#) 5
[Day report](#)
[Settings](#)



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► HISTORY Nothing recorded Show

► BEHAVIOUR & PLAY Friendly, Energetic · big dogs: does well · Daycare often Show

▼ VACCINES Hide

VACCINE	DATE GIVEN	EXPIRES	STATUS
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One setting changed.

Payments

Square Readers cannot be driven remotely — the Terminal API only pushes to Terminal and Register hardware. So this uses the **Point of Sale API**: the browser hands off to the Square app with the amount filled in, and Square returns to `/pay/return`.

The hand-off replaces the page entirely, so what the payment was for is written to `localStorage` first, keyed by a token Square echoes back. On return the write is made idempotent by transaction id — Square can return more than once, and a refresh must not double-record.

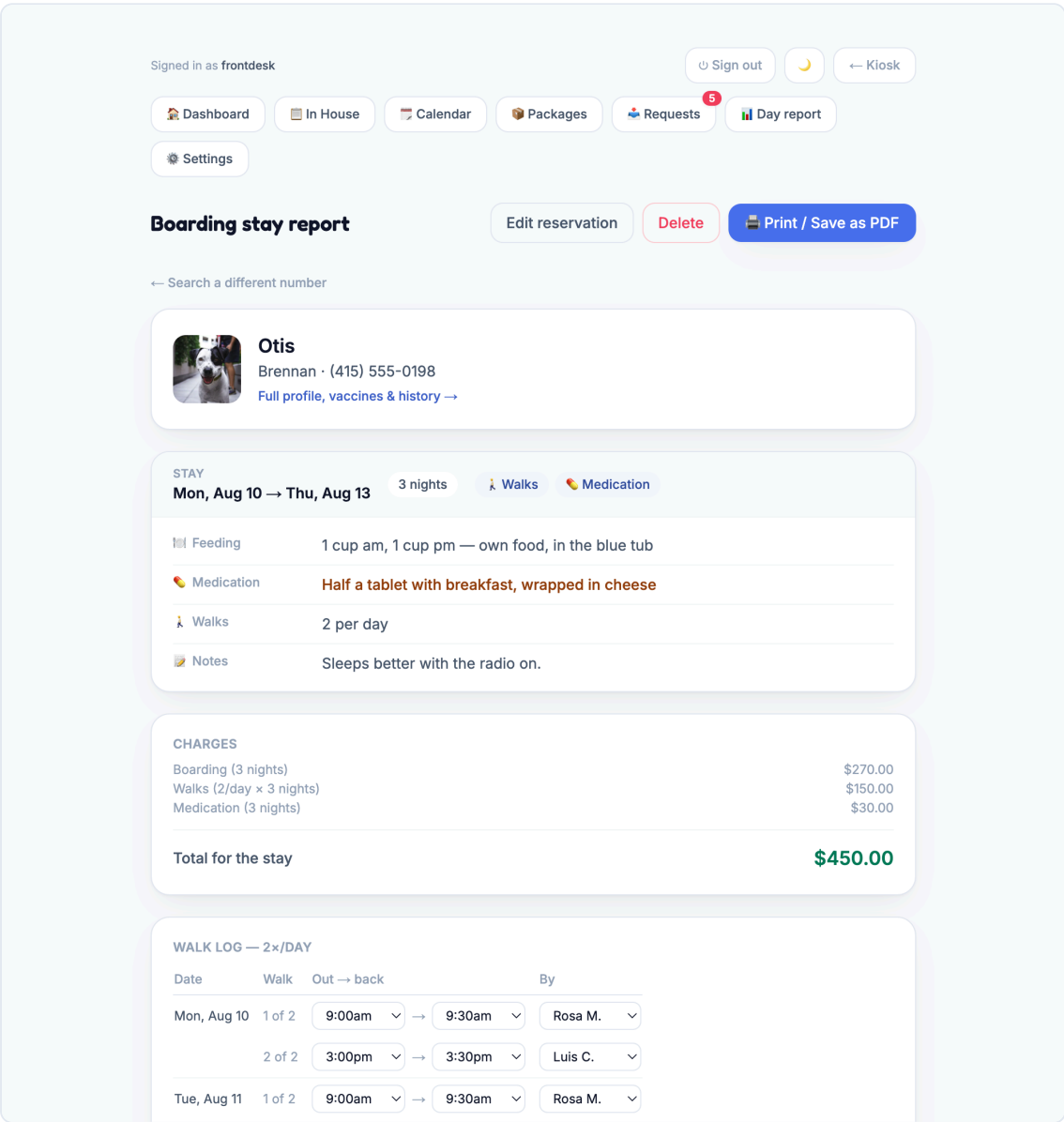
The failure that mattered most was the opposite one: a reload landed on "Paid, but not recorded" because the stash is consumed on first use. Staff would have entered the payment a second time. The orphan branch now checks the ledger before saying anything.

Security

- Supabase Auth sessions; no public sign-up
- RLS on every table — the UI gate is convenience, the policy is the boundary
- The kiosk holds its own session, because a locked-down database will not serve an anonymous caller either
- An idle re-lock on staff pages, which ends the convenience, not the session

Printing

Every operational list has a print stylesheet: interactive controls disappear, colour becomes rules and bands, and half-filled logs print as lines to complete by hand. Kennels still run on clipboards, and a screen is not much use in a wash room.



The stay report is designed to be printed as much as read.